Sangji Lee, M.S.

AI for Healthcare · Clinical NLP · Clinical AI Agents · AI Evaluation

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EDUCATION

Sungkyunkwan University *Mar 2024 – Feb 2026*

M.S. in Digital Health (Advisor: Prof. Won Chul Cha)

- GPA: 4.14/4.5
- Thesis: Automated KCD Coding Using Large Language Model with Integration of Coding Guidelines

Dankook University *Mar 2018 – Feb 2022*

B.S. in Public Health (Major: Health Care Administration)

- GPA: 4.2/4.5 (Cumulative) | 4.3/4.5 (Major)
- Relevant Coursework: Hospital Statistics, Health Information Technology, Health Data Management

PUBLICATIONS

Manuscripts Under Review

[1] **Lee, S.** & Cha, W. C. (2026). Matching Coding Functions to Computational Mechanisms: The CLEVER Framework for Automated Clinical Coding. *JAMIA Open* (revised manuscript under review).

[2] Kwon, Y.*, **Lee, S.***, Kim, J., Sul, J., Chang, H., Moon, H., Cha, W., Kim, T., & Choi, E. (2026). K-InsSim: Benchmarking LLM Agents on Evolving Inquiries in Health Insurance Consultation. Under review at *ACL Rolling Review* (October 2026 cycle).

*Equal contribution (co-first authors).

Peer-Reviewed Publications & Accepted Papers

[3] Kwon, Y., Kim, J., Choi, J., Rabaey, P., Kim, M., Im, S., Yang, J., Lee, J.-M., **Lee, S.**, Kim, J., Yoon, H., Kwon, H., & Choi, E. (2026). Towards Error-Free EHRs: Reasoning-Intensive Consistency Verification Between Clinical Notes and Structured Tables in Electronic Health Records. *NeurIPS 2026*, Evaluations & Datasets Track.

[4] Jang, E. H., Aguirre, J., **Lee, S.**, Moon, H., & Cha, W. C. (2025). Span-based annotation framework for LLM-based clinical named entity recognition: development and validation using Korean emergency department notes. *JAMIA Open*, 8(6).

[5] **Lee, S.** & Cha, W. C. (2025). Optimizing Automated KCD Coding: A Retrieval-Verification Approach. *Studies in Health Technology and Informatics*, 327, 872–873 (Proceedings of Medical Informatics Europe 2025; Oral Presentation).

RESEARCH EXPERIENCE

Healthcare Digital Transformation Research Institute

Researcher *Mar 2026 – Present*
Student Researcher (M.S. Program) *Mar 2024 – Feb 2026*
Intern Researcher *Oct 2023 – Feb 2024*

- **CLEVER: Guideline-Decomposed Framework for Automated Clinical Coding**
  - Lead Researcher | PI: Prof. Won Chul Cha
  - Designed a three-stage framework integrating guideline-aware LLM extraction, vector-based terminology retrieval, and rule-based correction to reflect expert coders' reasoning.
  - Achieved an exact-code F1 score of 0.77 on Korean emergency department notes.
- **K-InsSim: Multi-turn LLM Agent Benchmark for Health Insurance Inquiries**
  - Lead Researcher | PIs: Prof. Edward Choi, Prof. Taerim Kim
  - Developed a benchmark for evaluating tool-augmented LLM agents through multi-turn, simulated health insurance consultations and evidence-grounded decision tasks.
  - Designed case-based and real-query evaluation settings with evolving inquiries and time-sensitive reimbursement criteria.

- **RAG-based Health Insurance Review Decision Support Chatbot**
  - Lead Researcher | PI: Prof. Taerim Kim
  - Built a RAG chatbot grounded in Korean health insurance reimbursement criteria.
  - Conducted a proof-of-concept deployment with the Samsung Medical Center Insurance Review Team.
- **Clinical NLP Dataset Development: NER and PHI De-identification**
  - Research Assistant | PI: Prof. Won Chul Cha
  - Contributed to annotation guidelines and labeling schema design for clinical NER on emergency department notes.
  - Designed PHI de-identification annotation guidelines and managed dataset quality assurance for emergency department notes.

PROFESSIONAL EXPERIENCE

Department of Medical Records, Severance Hospital *Mar 2022 – Oct 2023*
Health Information Manager
Nationally licensed Health Information Manager (Republic of Korea)

- Performed diagnostic (KCD-8) and procedure coding (ICD-9-CM Vol. 3) with documentation deficiency analysis on 20–30 inpatient records daily across departments, maintaining low error rates verified through monthly audits.
- Managed DRG coding for inpatient cases, aligning assignments with classification criteria and coordinating with physicians on documentation revisions.
- Assigned cause-of-death codes through clinical notes review and consultations with physicians to clarify primary causes.

HONORS & AWARDS

Fulbright Graduate Student Program *Fulbright Korea, 2026*
Alternate Candidate

Honorable Mention, Healthcare Big Data & AI Competition *Health Insurance Review & Assessment Service, 2026*
Top 5% of 304 participating teams

Outstanding Research Achievement Scholarship *Sungkyunkwan University, 2025*

Dean's Award *College of Health Sciences, Dankook University, 2022*

Honors Graduate *Dankook University, 2022*

Academic Excellence Scholarship *Dankook University, 2018, 2019, 2021*

Entrance Merit Scholarship *Dankook University, 2018*

PATENT

[1] Kim, T., Park, C., **Lee, S.**, Hernandez, J. (2025). Method for Providing a Patient's Health Insurance Reimbursement Eligibility. Korean patent application No. 10-2025-0138818.

TEACHING EXPERIENCE

Teaching Assistant, "AI Introduction for Healthcare Services" *Sungkyunkwan University, 2024*

- Managed course logistics, including lecture preparation and exam administration.
- Coordinated student inquiries and assignment workflows.

LANGUAGES

English | Fluent (TOEFL 110/120) **Korean** | Native **Japanese** | Intermediate (JLPT N3)